

Test Report

Hostel Management System, version 1.1.0 | 08 October 2026 | OffbeatSoftwaresolutions

1. Result

319 / 319 automated test runs passed	3 of 3 full runs green on the same build	127 / 137 test cases passed by automated tests	0 open defects
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The application passed every automated test. No failures were found.

Of the 137 test cases in **TestCases.xlsx**, 127 were passed by the automated tests. For 15 of these the program logic is proven, and a short look on the client PC is still advised (for example a real Gmail delivery). The remaining 10 cases can only be checked on the client PC, such as installing and uninstalling. They are listed in section 5.

2. How the tests were run

- **Where:** GitHub Actions, Windows Server (windows-latest), with the Microsoft Access Database Engine 2016 (64 bit) installed exactly as on the client PC.
- **What:** the release build of the application, commit 9512db4. The same build produced the installer HostelManagement-Setup.exe.
- **How:** 249 test methods (319 runs including data rows). Every test creates its own new Access database, so tests never affect each other or real data. Screens are opened and clicked in a real Windows session.
- **Repeats:** the full set was run three times (runs 58, 59 and 60) to confirm the results are stable. All three passed: 957 of 957 runs.

3. Automated results per area

Area	Test methods	Runs	Passed	Failed
Agreement and beds	10	10	10	0
Amounts in rupees and sorting	4	9	9	0
Application Settings: database check	2	2	2	0
Attendance	6	6	6	0
Backup / Restore	10	10	10	0
Billing periods (half yearly, quarterly)	4	14	14	0
Colleges	12	13	13	0
Dashboard	5	10	10	0
Data access and SQL safety	12	12	12	0
Database setup and versions	13	13	13	0
Email	17	17	17	0
Email, phone and text checks	5	15	15	0
Empty database for the installer	1	1	1	0
Hostels	12	15	15	0
Invoices	11	13	13	0

Main window and menu	9	9	9	0
Parents	7	7	7	0
Payments and receipts	12	23	23	0
Pending Dues	9	9	9	0
Reports (PDF and Excel)	9	9	9	0
Room Allocation	15	15	15	0
Rooms and rent	25	28	28	0
Screens drawn at 1280 x 760	3	3	3	0
Services and transport	11	11	11	0
Sign in and Admin Account	7	13	13	0
Students, photos and documents	18	32	32	0
Total	249	319	319	0

4. Test case coverage per module

Module	Test cases	Passed (automated)	Passed, quick look advised	Manual on client PC
Installation	6	2	1	3
Sign in	6	6	0	0
Admin Account	6	6	0	0
Hostels	5	5	0	0
Colleges	3	3	0	0
Rooms	9	9	0	0
Services	3	3	0	0
Students	11	10	0	1
Parents	3	3	0	0
Room Allocation	9	8	0	1
Agreement	7	6	1	0
Attendance	9	6	2	1
Invoices	11	10	1	0
Payments	9	8	1	0
Pending Dues	7	5	2	0
Email	9	5	3	1
Dashboard	4	3	1	0
Reports	7	6	1	0
Backup / Restore	7	5	2	0
Application Settings	2	1	0	1
General	2	2	0	0
Uninstall	2	0	0	2

Total	137	112	15	10
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The Status, Tested by, Date and Remarks columns of TestCases.xlsx are filled in, and each passed case names the automated tests that proved it. The new sheet "Automated Run" holds the run details.

5. Checks to do on the client PC

These 10 test cases need the real PC, the installer or the real Gmail account, and cannot be run automatically:

ID	Module	Test case	What to do
TC-001	Installation	Install on a new PC	Installer: run on the client PC
TC-002	Installation	Access Database Engine missing	Installer: run on a PC without the engine
TC-006	Installation	User guide shortcut	Start menu shortcut on the client PC
TC-047	Students	Search and filters	Type in the search box and use the filters on the Students screen
TC-061	Room Allocation	Agreement offered after check-in	Question box after check-in
TC-074	Attendance	Unsaved changes	Question box when leaving unsaved changes
TC-106	Email	Wrong app password	Needs the real Gmail account
TC-133	Application Settings	User guide	Click User Guide in Application Settings
TC-136	Uninstall	Uninstall keeps data	Uninstall on the client PC
TC-137	Uninstall	Reinstall uses old data	Reinstall on the client PC

For these 15 cases the logic passed, but a quick look on the client PC is still advised:

ID	Module	Test case	Still check by hand
TC-005	Installation	Update keeps data	Run the new setup over the installed version once
TC-062	Agreement	Create agreement PDF	Open the PDF once to look at the layout
TC-076	Attendance	Email parents of absent	Real delivery through Gmail
TC-077	Attendance	Email again	Question box before emailing again
TC-086	Invoices	View / Print and Save PDF	Print once on the office printer
TC-097	Payments	Filter and search	Search box and filters on the Payments screen
TC-102	Pending Dues	Send reminder	Real delivery through Gmail
TC-103	Pending Dues	Remind all overdue	Real delivery through Gmail
TC-105	Email	Test email	Needs the Gmail app password, entered on the client PC
TC-108	Email	Email invoice	Real delivery through Gmail
TC-109	Email	Email receipt	Real delivery through Gmail
TC-115	Dashboard	Chart	Hover over a bar to see the amount
TC-123	Reports	Export Excel	Open the file once in Excel

TC-129	Backup / Restore	Restore	Program restart after restore
TC-130	Backup / Restore	Undo restore	Restore the 'Before a restore' backup once

The End-to-End Workflow sheet (26 steps) is the best way to do these checks in one sitting. Each step names its related test cases.

6. Problem found and fixed during testing

Rare program crash inside the Access Database Engine. Now and then the test process ended without an error message. A crash dump showed the cause: the Access Database Engine unloads when the last connection to the database closes, and loading it again occasionally fails.

Fix: the application now keeps one database connection open from start to exit, on its own thread. It is closed only while a backup is restored. This also protects the installed program when it is left open and idle for a long time.

Result: three full runs passed afterwards with no crash.

7. Notes for go-live

- **Gmail:** email sending was tested with a test mail server. Real delivery needs the 16 letter Gmail app password, entered once in Email Settings on the client PC. Then use Send Test Email (TC-105).
- **Windows SmartScreen** may warn on the first install because the installer is not code signed. Choose More info, then Run anyway.
- **Data:** keep the daily automatic backup on, and set the second backup folder to a USB drive or cloud folder.
- **Future updates:** if a later version changes the database layout, it must include an upgrade step for databases that already hold data. Please take Backup Now before installing any update.

Run links: <https://github.com/offbeatsoftware/hostelmanagement/actions/runs/37736840117> · [/37736847211](#) · [/37736850027](#)